

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Mechanical Engineering

2. Subject Name: Thermal Engineering Lab
Credits: 2

L-T-P: 0-0-3

3. Pre-requisites: NIL

4. Syllabus and reference books:

Syllabus:

IC Engine Lab.

1. Dismantling and Assembly of four-cylinder engine
2. Measurement of Engine Performance
3. Calibration of SI engine
4. Optimization of fuel injection of diesel engine on performance and emissions

Refrigeration and Air-Conditioning lab.

- 1 a) Study on refrigerant compressors
b) Study on indoor thermal comfort
2. Steady state measurements on a vapour compression refrigeration test-rig
3. Pull-down and cyclic characteristics of a small cold storage
4. Performance of a summer air conditioning system

Basic Combustion

1. Measurement of Flame speed
2. Heat transfer in fluidised bed
3. Heat transfer during cold air mixing with hot exhaust gas
4. heat transfer characteristics of a fin and tube radiator

Reference Books:

NA

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	IC Engine Lab. 1. Dismantling and Assembly of four-cylinder engine 2. Measurement of Engine Performance 3. Calibration of SI engine 4. Optimization of fuel injection of diesel engine on performance and emissions	12

2.	Refrigeration and Air-Conditioning lab. 1 a) Study on refrigerant compressors b) Study on indoor thermal comfort 2. Steady state measurements on a vapour compression refrigeration test-rig 3. Pull-down and cyclic characteristics of a small cold storage 4. Performance of a summer air conditioning system	12
3.	Basic Combustion 1. Measurement of Flame speed 2. Heat transfer in fluidised bed 3. Heat transfer during cold air mixing with hot exhaust gas 4. Heat transfer characteristics of a fin and tube radiator	12
Total number of hours		36